

Arshia Fazeli

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SUMMARY

PhD candidate in Chemical Engineering with 5 years of experience developing high-fidelity computational models and numerical solvers for multiphysics problems. Proven ability to design, implement, and verify complex software systems in Python (NGSolve, PETSc) and apply advanced numerical methods — discontinuous Galerkin FEM, finite-volume methods, adaptive mesh refinement — to real engineering problems. Published researcher with strong skills in scientific software engineering, data analysis, and machine learning pipelines.

EDUCATION

University of Waterloo

Sept 2021 – Sept 2026 (Expected)

PhD in Chemical Engineering, Waterloo, ON, Canada

- GPA: 92.67/100
- Dissertation: Well-posed two-fluid models for dispersed multiphase flows (Advisor: Nasser M. Abukhdeir)
- Key courses: Computational Fluid Dynamics, Multiphase Flow Dynamics, Advanced Engineering Mathematics, Transport Phenomena

University College London

Sept 2018 – June 2021

BEng in Chemical Engineering, London, UK

- First Class Honours
- Key courses: Computational Modelling and Analysis, Reaction Engineering, Process Dynamics and Control, Particulate Systems and Separation Processes

SKILLS

Programming Languages: Python, MATLAB, C++

Scientific Computing: NGSolve, PETSc, OpenFOAM, Ansys CFX, ParaView, GMSH, NETGEN

Python Libraries: NumPy, Numba, scikit-learn, pandas, Matplotlib, SymPy, unittest, pytest

Tools & Workflow: Git, $\text{\LaTeX}$, Linux, LAMMPS, PLUMED

RESEARCH EXPERIENCE

University of Waterloo

Sept 2021 – Present

Computational Multiphysics Researcher, Waterloo, ON, Canada

- Formulated a well-posed Two-Fluid Model (TFM) incorporating a laminar dispersion force closure to broaden the hyperbolicity domain for dispersed gas–liquid and liquid–liquid multiphase flows.
- Developed discontinuous Galerkin (DG) finite element solvers in Python/NGSolve for coupled multiphysics systems: two-fluid models, incompressible Navier–Stokes, Cahn–Hilliard equation, and monolithic Cahn–Hilliard Navier–Stokes (CHNS) two-phase flow.
- Implemented adaptive mesh refinement driven by a Zienkiewicz–Zhu gradient-recovery error estimator, enabling targeted refinement in regions of large solution gradients.
- Verified solver accuracy via the method of manufactured solutions (MMS), quantifying convergence rates and discretization error.
- Validated models against published experimental data (Pfleger et al. 1999 bubble column; Rodriguez et al. 2012 liquid–liquid pipe flow).

University College London

Aug 2020 – Dec 2020

Research Associate, London, UK

- Conducted molecular dynamics simulations using LAMMPS and PLUMED to investigate nucleation pathways in colloidal suspensions.

TEACHING EXPERIENCE

University of Waterloo

Sept 2021 – Present

Teaching Assistant, Waterloo, ON, Canada

- Led tutorials and office hours for graduate and undergraduate courses: Theory and Application of Transport Phenomena (CHE601), Introduction to Computational Methods (NE113), Continuum Mechanics (NE318), Computer Literacy & Programming (NE111/CHE120).
- Engineered a randomized Python `unittest` autograder for a programming course — freshly generated test inputs each run prevent hardcoded solutions; weighted scoring, safe execution via `runpy`, and per-student filtered error feedback.

SELECT PROJECTS

LLM-Generated Code Detector

- Built an end-to-end ML pipeline detecting AI-generated student code: engineered 18 AST-based and stylistic features (Halstead metrics, Shannon entropy, comment semantics, call nesting depth) and trained a Random Forest classifier via 5-fold stratified cross-validation with `RandomizedSearchCV`, achieving ROC-AUC of 0.95.
- Implemented a TF-IDF cosine similarity plagiarism checker to flag near-identical submission pairs across a student cohort.
- *Tools: Python, scikit-learn, pandas, NumPy*

DG Solver for Euler–Euler Two-Fluid Model

- Developed a DG finite element solver for simulating dispersed multiphase flows using the two-fluid model, with adaptive implicit time-stepping and mixed HDiv/L2 function spaces.
- Verified via the method of manufactured solutions and validated against the Pflieger et al. (1999) bubble column and Rodriguez et al. (2012) liquid–liquid pipe flow experiments.
- *Tools: Python, NGSolve, PETSc, ParaView*

DG Solver for Cahn–Hilliard Navier–Stokes System

github.com/arfazeli/CHNS_DG_solver

- Implemented a monolithic DG solver for the coupled Cahn–Hilliard Navier–Stokes system, simulating the evolution of two immiscible incompressible fluids separated by a diffuse interface.
- *Tools: Python, NGSolve*

DG Solver for Cahn–Hilliard Equation

github.com/arfazeli/CH_DG_solver

- DG solver for the Cahn–Hilliard phase-field equation capturing spinodal decomposition dynamics of binary fluid systems.
- Implemented adaptive mesh refinement driven by a Zienkiewicz–Zhu gradient-recovery error estimator on ∇c , enabling targeted refinement at the diffuse interface.
- *Tools: Python, NGSolve, NETGEN, GMSH, ParaView*

Steady Incompressible Navier–Stokes Solvers with Scalar Transport and Reaction

[SIMPLE](#), [SCGS](#)

- Implemented FVM solvers for the 2D incompressible Navier–Stokes equations using SIMPLE and SCGS velocity–pressure coupling; extended the SIMPLE solver to include scalar transport and reaction source terms.
- Implemented multiple convection and diffusion discretization schemes; leveraged Numba JIT compilation for significant runtime speedups.
- *Tools: Python, NumPy, Numba, Matplotlib, SymPy*

Euler Equations Solver Using Flux Vector Splitting

github.com/arfazeli/1D_Euler_FVS

- Implemented a finite-volume solver for the 1D compressible Euler equations using Steger–Warming, Van Leer, and AUSM+ flux splitting schemes; the AUSM+ scheme uses an advection/pressure operator-split formulation for improved accuracy on contact discontinuities and shocks.
- *Tools: Python, NumPy, SymPy, Matplotlib*

CFD Simulation of a Side-Dump Combustor with Variable Inlet Angles

- Performed CFD simulations of turbulent airflow in a side-dump combustor; validated against experimental velocity and turbulence data and investigated the impact of side-inlet angle on fuel–air mixing.
- Conducted mesh refinement and error analysis to quantify discretization and modelling errors.
- *Tools: Ansys CFX*

LEADERSHIP AND SERVICE

Chemical Engineering Graduate Student Association (CEGSA), University of Waterloo <i>VP Academic</i>	<i>Jan 2022 – Jan 2023</i>
UCL Technology Society <i>Welfare Officer</i>	<i>Aug 2020 – July 2021</i>
Engineers Without Borders <i>Outreach Volunteer</i>	<i>Oct 2019 – Apr 2020</i>
UCL Project Impactive <i>Engineer</i>	<i>Oct 2019 – Jan 2020</i>

PUBLICATIONS

- [2] **A. Fazeli**, S. Rhebergen, N. M. Abukhdeir. “Canonical formulation of two-fluid model with enhanced physical fidelity.” *International Journal of Multiphase Flow* (Under Review, 2025). [Preprint](#)
- [1] **A. Fazeli**, S. Rhebergen, N. M. Abukhdeir. “Laminar dispersion force effects on two-fluid modelling and simulation of bubble column hydrodynamics.” *International Journal of Multiphase Flow*, 2026. [DOI](#)

AWARDS

CSCHE 2024 Best Poster Award	Nov 2024
Garry L. Rempel Memorial Graduate Scholarship , University of Waterloo	May 2022
Best Final Design Project Team Award , University College London	June 2021